Transgender voice training assistant based on ML classifiers and algorithms

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Abstract

This project designed and implemented an application for assisting voice training in transgender voice therapy by visualizing some vocal features and providing informative analysis of these features. To design this application, we need to answer three questions, which are what algorithms to be used for distinguish between masculine and feminine voice; what vocal features are informative for voice therapy; how to optimize our application and improve its scalability? Correspondingly, we modified a powerful Convolutoinal Neural Network (CNN)-based speech gender detection model, extracted useful vocal features to perform statistical analysis, and adopted some optimization tools and a distributed computing system. The experimental results showed that our application is successful in providing numerical analysis in vocal gender features throughout the modified CNN model and statistical methods, as well as computing structure with low-latency for users' requests.

1 Introduction

Danya Lagos' study [18] shows that voice is an important aspect of an individual's gender expression. Transgender populations are suffering from the inconsistent gender classification by others, and this disadvantage brings serious mental health disabilities, such as poor self-rated health condition [18]. In this sense, the quality of vocal training becomes a social demand for transgender populations. Changing the perceived gender classification is an important step in the transition process. We aim to provide an accessible (i.e. requires no extra personnel) and objective (i.e. provide a detailed analysis of their voice) tool for people undergoing voice training, helping them get closer to their goal.

In general, automatic classification of perceived gender of an individual's voice is done by extracting different features of the voice. There are three features that commonly used in vocal gender classification task, pitch, formant, and harmonics [26]. In the formal definition,

- **Picth**: the relative highness or lowness of a tone as perceived by the ear[4].
- **Formant**: the broad spectral maximum of a person's acoustic resonance [29].
- **Harmonics**: a component frequency of an oscillation or wave in the voice.

The current popular gender classification tools are limited by their output value, such as inaSpeechSegmenter [12] is using a simple binary output. Our approach is to provide a program can not only estimate the confidence of the gender classification, but also generate a detailed analysis to guide the users to make progress in their training.

In our work, we will focus on three research questions at this stage.

1) RQ_1: Can we use algorithms to classify and distinguish between masculine and feminine voice?
2) RQ_2: What vocal features can be used to distinguish between masculine and feminine voice? How are the vocal features distributed among genders? How can we visualize the vocal features to provide optimal feedback for changing the voice?
3) RQ_3: How can we run the algorithms as efficiently as possible, and scale the computation to provide a fast, responsive service to our users?

2 Related Work

2.1 CNN-based gender classifier

InaSpeechSegmenter [12] is a python library that contains a pretrained model that takes voice as input, and returns binary gender classification, which is either male or female. InaSpeechSegmenter [12] shows the reliability of machine learning algorithm on gender calssification, rather than the traditional stastics tools.

2.2 Short-Time Fourier Transform

Short-time Fourier transform (STFT) [7] reads the voice input, and generates a time-frequency distribution of the change of signals. STFT divides the long input into short equal length intervals, it tells the relationship between the amplitude versus time and frequency on any source of signals.

2.3 Pitch vs Formant Positional Graph

The program done by TransVoiceTech [30] provided an interesting visualization of the geometric information for vocal gender on a pitch versus formant positional graph. The graph generated from [30] could guide users to have an understanding on their vocal characteristics, specific to picth and formant. Users could train to adjust their picth and formant to achieve their desired target vocal gender.

3 Research Questions and Results

3.1 RQ1: Can we use algorithms to classify and distinguish between masculine and feminine voice?

3.1.1 ***Motivation*** In the voice training process, one major barrier is that people often cannot obtain objective feedback of how their voice sound. Since people obviously cannot hear their own voice, they cannot assess their voice on their own. Also, playback of a recording of one's voice is very far from how their voice sound to other people, since most affordable microphones and speakers distort the voice in the analog encoding process. Often, one might rely on their friends or others in the community to provide feedback for their voice. However, few people have close friends who are always around, and they are very likely biased toward giving a affirmative feedback rather than constructive criticism. Also, it requires great courage to practice new voicing techniques in front of their friends. Therefore, we need an automatic classification model serving as an objective friend always around to listen to one's voice and provide feedback.

InaSpeechCNN [12] is a powerful speech gender detection tool based on a deep CNN architecture. It first segments audios and then outputs gender classification results for each segment. This model is shown to outperform Gaussian Mixture Model and I-vector gender segmentation system in a large French vocal dataset. However, the basic InaSpeechCNN only provides binary outputs which tell users their voice sounds female or male. Since our project aims to provide numerical analysis on gender features from users' voice, we modified the outputs of InaSpeechCNN to obtain the probabilities of how much the voice sounds feminine or masculine.

3.1.2 ***Methodology*** To generate numerical gender features from vocal data, we first obtained the raw classification model output from each segment of an audio. Each output is consist of two probabilities corresponding to how much this segment audio sounds feminine or masculine respectively. Then we calculated the average of these outputs. This value is taken as the final numerical gender feature of this audio.

3.1.3 ***Evaluation and Results*** We evaluated the modified model on three large-scale datasets, which are CN-Celeb [14], VoxCeleb1 and VoxCeleb2 [21]. The first dataset is of Chinese utterances, while the other two are English. The size and gender distribution of each dataset is shown in Figure 1 .

The modified model was tested against users. If a user has many audio records in the testing dataset, we will collect the model outputs of each of his/her records, and take the mean of these outputs as the final one for this user. However, to evaluate the basic classification performance of this model on these languages, we still need to map the final model output for each user into a binary label, which is female or male, according to a threshold of 0.5. If the model output is not less than 0.5, then we say this user sounds feminine. The model evaluation results are shown in Table 1 and 2.

As we can see from these two tables, for each gender and in every dataset, our modified InaSpeechCNN model achieved relatively high precision and recall rates, which supports the use of this model in our final application.

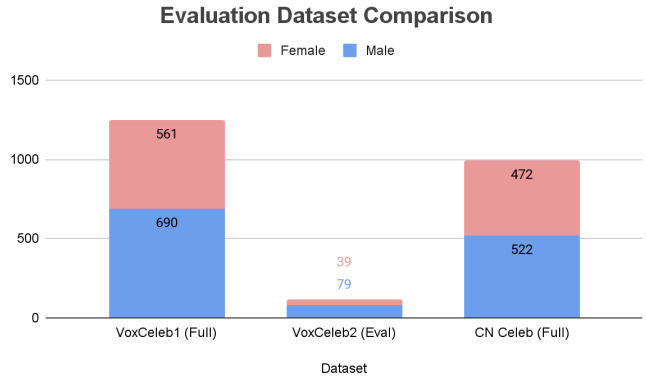


Figure 1: Information on Evaluation Datasets

Dataset	CN-Celeb	VoxCeleb1	VoxCeleb2
precision	95.32%	98.93%	100%
recall	99.6%	98.93%	92%

Table 1: Evaluation Results on Female Utterances

Dataset	CN-Celeb	VoxCeleb1	VoxCeleb2
precision	99.59%	99.13%	96%
recall	95.69%	99.13%	100%

Table 2: Evaluation Results on Male Utterances

3.2 RQ2: What vocal features can be used to distinguish between masculine and feminine voice? How are the vocal features distributed among genders? How can we visualize the vocal features to provide optimal feedback for changing the voice?

3.2.1 ***Motivation*** Even though machine-learning models can effectively and efficiently classify perceived gender, the classification results cannot be explained due to the black box nature of machine-learning models. To provide feedback for the users to improve their voice and sound closer to their target voice, we need an explainable method of classification, with statistical analysis on vocal features that can be extracted and changed through altering voicing techniques.

3.2.2 ***Approach*** The perceived vocal pitch, measured by the fundamental frequency, is commonly believed to be the major difference between feminine and masculine voice. However, vocal tract resonance measured by different formant frequencies are also important in gender perception [16] [25]. A mismatch between pitch and resonance will often result in unnatural-sounding voice. In addition, spectral tilt, measured by the difference in harmonics, represents the breathiness of the voice, which might also show a difference across genders. Wright noted that a steep spectral tilt

signifies a breathy voice, and a shallow voice denotes a creaky voice [31].

To extract these vocal features, we compared the running time of existing algorithms and implementations, and selected the implementation with high enough accuracy and fast response time. The detailed comparison will be discussed in the optimization section. We estimated fundamental frequency using the auto-correlation method proposed by Paul Boersma [8], implemented in Praat [9], and adapted to python by Jadoul et al [17]. We estimated formant based on the linear predictive encoding method proposed by John Burg, and spectral tilt is computed through taking the difference in harmonics, with the algorithm from a tool written by M.J. Owren [24].

To gather the statistical distribution of the vocal features, we computed the vocal features across all 148642 audio files available in the VoxCeleb1 dataset [21], and gathered the mean of each person. Then, we gathered the means of all people of each sex for each feature, which gave us a total of 10 sets of means (since there are 5 features, and our dataset only consists of people of two genders). We then computed statistical distribution for each of the sets of means.

In order for users to grasp how each of their vocal features compares to their target genders, we used kernel density estimation by Epanechnikov [13] to calculate the probability of how likely user's voice falls into each gender. For each vocal feature, the percentage calculation is done by first calculating the mean across the entire audio that the user uploaded, then dividing the estimated density for feminine voice by the total estimated density. As seen in Fig 2, We also show the position of the user's mean relative to the distributions on each feature.

For a more abstract view, we implemented the coordinate chart designed by Zheanna [30] to give user an overview of their voice as seen in Fig 3. This plot might be more intuitive and compact than the kernel density distribution graphs, since it combines the results of all three distributions (pitch, f1, and f2). The X position of the point shows the perceived vocal gender according to the combination of formant values (f1 and f2), and the Y position relies on the pitch. This chart allows users to easily identify whether they need to improve on pitch or resonance, and it also enables the users to determine whether their voice sounds natural or unnatural– while the top right and bottom left centers represents feminine and masculine averages, the top left and bottom right areas represent voices that might be perceived as "unnatural." For example, during the voice training process, one might have a high voice with a masculine undertone (high pitch, low resonance), or a low voice with feminine resonance. Seeing this chart will intuitively inform them where to improve.

The coordinate is computed with a piecewise function for each axis, with the center coordinates being the means of each gender. For example, on the Y axis, 0.25 represents that the average pitch of your recording equals the mean pitch for masculine voice, and 0.75 represents mean pitch for feminine voice. Then, the border values represents the 1st percentile and the 99th percentile of the values. The position is determined by a linear piecewise function between these four values.

In addition to visualizing statistical features, we displayed a spectrogram of the voice along with frequency overlays. Spectrograms are used extensively in the field of speech processing [15], which combines frequency space with time space to provide an overview of the sound's frequency peaks across time. We used short time Fourier transform (STFT) and a mel-filterbank to generate the mel-spectrogram [11]. We reshaped the Y-axis to the mel scale rather than displaying the spectrogram in linear scale because mel scale is considered to be a more accurate mapping of human's perception of different frequencies [28]. There are many different mel scale formulas proposed in history, we used the most popular O'Shaughnessey's formula because it is the simplest to compute [23]. We then overlayed the curves of the frequency features (pitch and formant) extracted by previously mentioned methods on top of the mel-spectrogram, as seen in Fig 4.



Figure 2: An example of statistic analysis

3.2.3 ***Evaluation and Results*** We are able to extract Features of fundamental frequency (a.k.a f0), f1, f2, f3, and spectral tilt from different genders. Overall, female voices features a higher frequency in all formants. Adopting different voicing techniques can allow the user to shift each of the formant frequencies independently or simultaneously in their voice [1], therefore, identifying their current formant frequency will help them select a particular voicing technique.

As seen in Fig. 6, female voices have a steeper spectral tilt, thus more breathy. But we can also see a major overlap between two groups, while this might be deterministic for identifying gender, it can provide a insight of whether the users' voice is too extreme to sound natural.

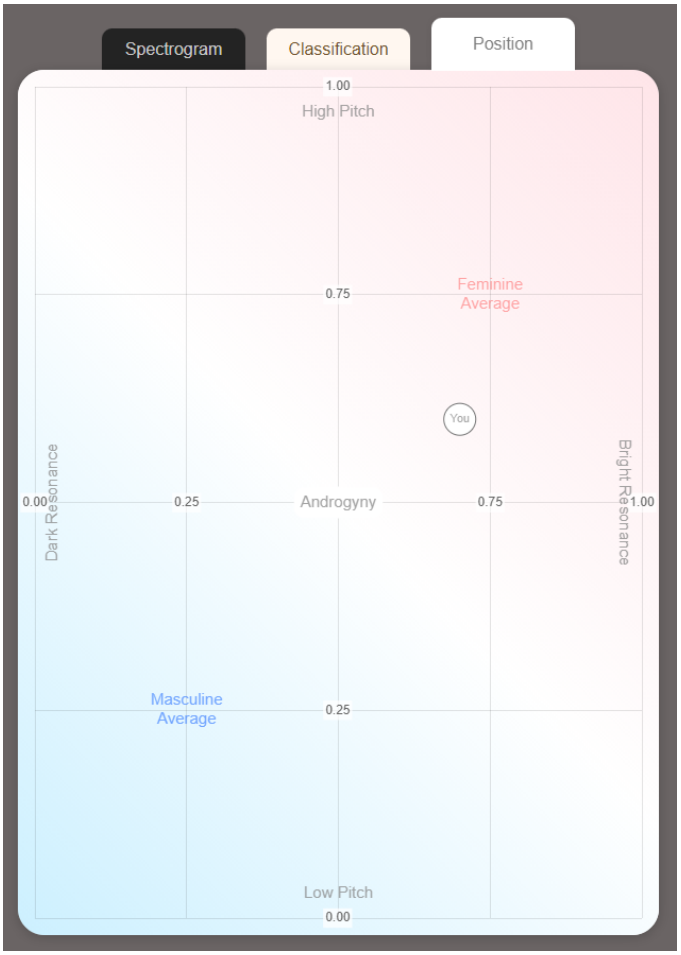


Figure 3: An example of a coordinate chart

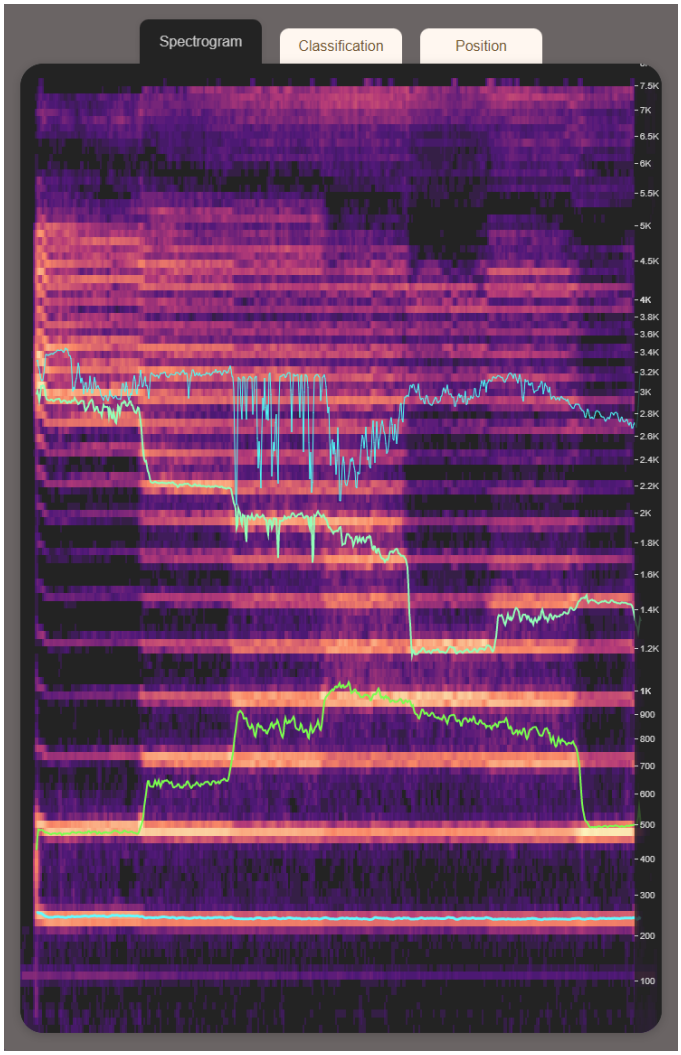


Figure 4: Mel Scaled Spectrogram, while lines represent different features

3.3 RQ3: How can we run the algorithms as efficiently as possible, and scale the computation to provide a fast, responsive service to our users?

3.3.1 ***Motivation*** Even though our first prototype is effective in providing visualization, feedback, and classification, the entire computation takes too long to be bearable for a real-world user, taking 18.17s to complete for a 70s audio file on a high-performance modern workstation, and crashes on an average gaming PC that a normal user might have. Therefore, we optimized our algorithms and created a distributed computing framework that allows volunteers to donate the resting computation power on their personal computers.

3.3.2 ***Optimization Approach*** To standardize running time measurements, all measurements used the same 70s feminine voice recording with a 48,000 Hz sample rate, computed on the same high-performance workstation, with an Intel E5-2696 v3 CPU, an NVIDIA GTX 1070 graphics card, and 128GB of DDR3-1866 RAM.

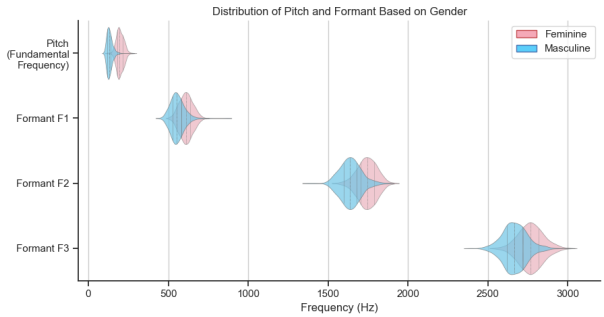


Figure 5: Distributions of vocal features based on gender.

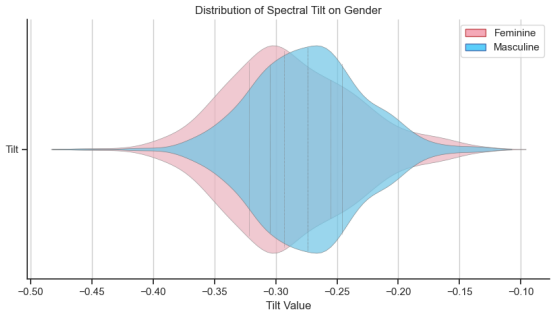


Figure 6: Distributions of spectral tilt based on gender.

The running time of each implementation is calculated by averaging the running time of 10 iterations under similar system load.

First, many different authors' algorithms complete the same underlying functions required in our program. These algorithms often have similarly accurate outputs even though their running time differs extremely. Also, different implementations of the same algorithm can also lead to significantly different running time because of their internal optimization. Therefore, the first step we took in optimization is comparing running time between existing implementations.

In the mel-spectrogram visualization, we initially used Tensorflow-JS [27] for completely frontend-based STFT and mel-filterbank calculations. However, after measuring the running time of several different STFT calculation libraries, results show that Tensorflow-IO [6] (28ms), a python library running on the backend, runs 99.93x faster than Tensorflow-JS (2798ms). One possible explanation is that, while Tensorflow-IO can take advantage of executing native machine-code bindings from Python, Tensorflow-JS is limited by JavaScript's object-based framework with heavy object wrapping/unwrapping overhead. And, although both frameworks utilize GPU resources, Tensorflow-IO can utilize NVIDIA CUDA's GPU parallel computing framework directly, while Tensorflow-JS can only utilize GPU through simulating WebGL[5] shader computation, adding many layers of complication[22].

Secondly, for algorithms we implemented in Python that don't have existing optimization, we reduced running time by rewriting Python code in compatible format and compiling python code into native machine-code using Numba's Just-In-Time LLVM compiler[2]. However, only pure Python code and a selection of Numpy functions are currently supported by Numba, and many function require extensive modifications before successful compilation, therefore we focused only on computationally heavy functions taking greater than 100ms.

In the inference step of the CNN model, we used the Viterbi decoding algorithm originally implemented by PyAnnote [10]. The original decoding algorithm took 389ms for our 70s audio. After statically typing all Numpy array creation and rewriting several unsupported Numpy functions, we were able to compile the decoder and its associated helper functions into machine code. After compilation, the same operation took only 24ms to execute, leading to a 16.2x performance increase for this step.

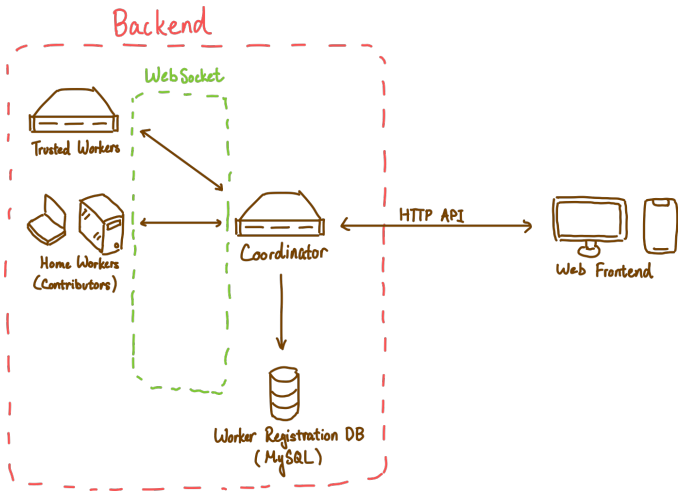


Figure 7: The Distributed Structure for Scaling

In the feature extraction step of the CNN model, we extracted log energy signal and mel-spectrogram with a triangular filter bank, originally implemented by SideKit[19]. The entire feature extraction step takes 113ms to compute. Since we have previously concluded that Tensorflow is the fastest library for computing STFT, we modified Tensorflow's STFT calculation to include log-energy signal output. We also modified SideKit's triangular filter bank for binary compilation. As the result, the new feature extraction function only took 15ms, leading to a 7.5x performance increase for this step.

Combining all optimizations, the total response time decreased from 18.17s to 7.82s for the 70s audio file.

3.3.3 ***Scaling Approach*** To improve the scalability of our application, we designed and implemented a distributed computing structure inspired by [20] in our application backend. It allows users to contribute their available computing resources only by running a program. A coordinator, workers, and a worker registration database are three core components in this structure. They are shown in Figure 7 . The coordinator is mainly responsible for registering workers in the database, scheduling tasks to workers, monitoring workers' authenticity, and interacting with users from the web frontend. Each worker here processes tasks assigned to it, and the registration database stores information of these workers. However, the main challenge to implement this structure is how to make sure workers are not modified, which means we need to ensure the authenticity of computation results from each worker, as well as their computing benchmark. For the first part of this challenge, we randomly selected responses from a worker for verification and ban a worker if its result does not match with the trusted worker's. While for the second part, we first let the coordinator send some random audio files to the worker to be testified, and this worker will return the corresponding results to the coordinator. If the coordinator finds these results are matched with those from the trusted worker, this worker will be registered into the database with its response time as its computing benchmark; otherwise, it will be rejected from being registered into the database.

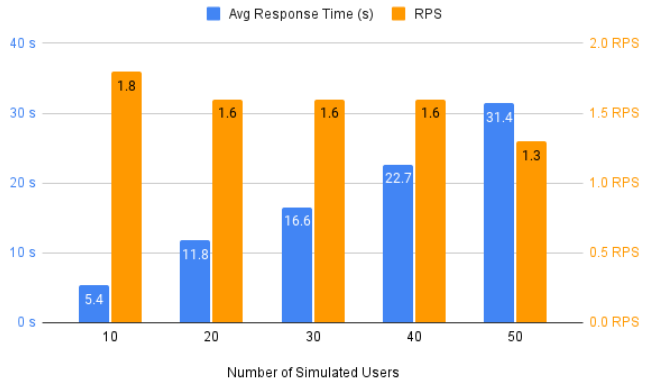


Figure 8: Stress Testing Results of a FastAPI Server

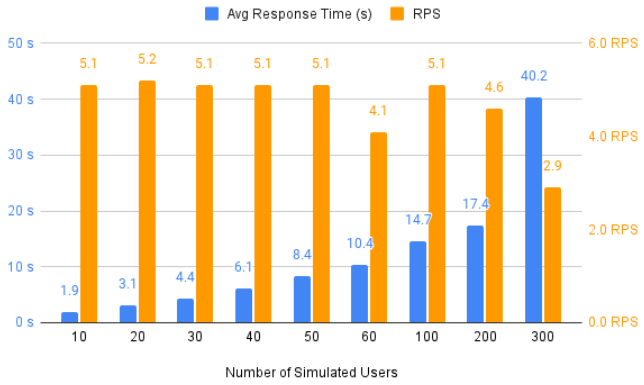


Figure 9: Stress Testing Results of a Distributed Structure

3.3.3.1 *Evaluation and Results*

To test the scalability of our distributed computing structure, we adopted *Locust*, an open-source server load testing tool [3]. Stress testing experiments are performed and compared between a FastAPI asynchronous server and our distributed system. The single-node FastAPI server is set up with a GTX 1070, having a CUDA Geekbench score of 45555. For building a distributed system, we used a GTX 1070, RTX 3060Ti, and GTX 1050M. This setting is equivalent to a total CUDA Geekbench score of 188453. The CUDA Geekbench score is proportional to how well a graphics processor performs machine-learning related computations.

We used programs to simulate users of our application and the simulated users repeatedly sent one request after another. The rate which users could send requests is measured by Request Per Second (RPS). Then, we recorded response time from the server to each user and took the average of these records. Figures 8 and 9 respectively illustrate the performance of the single server and the distributed system under stress testing. In both computing structures, the average response time increases with the number of users. However, when we have no more than 50 users, the distributed system consistently takes less time to respond, even though it handles requests nearly five times as many as those to the single server. A more straightforward comparison of these two structures is shown in Figure 10. When the distributed computing cluster has three nodes, its average response time is always less than 10 seconds as the number of users rises from 10 to 50, by contrast, the time a single server takes to respond increases from about 5 seconds to over 30 seconds. Also, from 20 to 50 users, the response time decreases by 3.75x on average, which is comparable to the 4.13x increase in total Geekbench score, with only 9.7% loss probably due to communication overhead. These results illustrated that our distributed computing structure does well in dealing with heavy workload, and scales without much overhead.

4 Threats to Validity

Formant estimation is also affected by the vowel the person is saying. With the lack of transcript of the data, the formant may be skewed towards certain vowel. In the future, we can incorporate

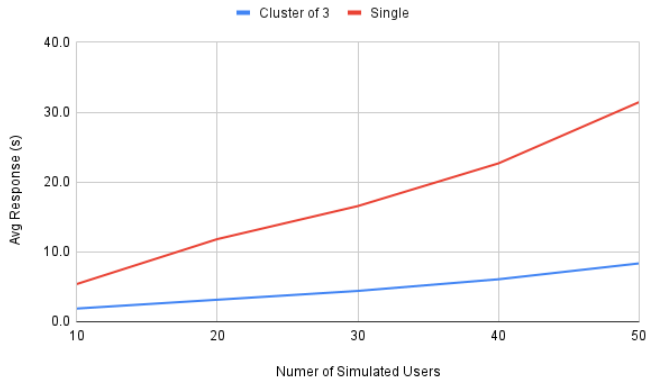


Figure 10: Comparison of Two Computing Structures

the probability distribution of vowels of English language to give a more accurate estimation of each voice sample.

Our original model is trained on french speakers' data, therefore a bias may be introduced due to different speaking habits between different language speakers, however, our model still achieved a high precision and recall, the influence may not be noticeable.

5 Conclusions

Our program successfully demonstrates the collaboration of machine learning algorithm and statistic analysis tools can work well on voice gender classification tasks. The confident output from the program overcame the limitation of binary classification. The visualization tool can additionally guide users to have a crystal clear understanding about the gap between the classified gender of their voice and their desired target gender. Our work would be beneficial to the transgender community, and we believe the users could get a more effectively aids from our program.

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